Signal Intelligence, OSINT, and Maritime Domain Awareness: A Technical and Strategic Analysis of the AIS Protocol, Dark Vessel Evasion, and Real-Time Tracking Schemas in Conflict Environments

The maritime domain serves as the primary artery of global commerce, and its security is increasingly dependent on the transparent exchange of signal intelligence. At the heart of this transparency is the Automatic Identification System (AIS), a sophisticated maritime communication protocol defined by the International Telecommunication Union (ITU) and mandated by the International Maritime Organization (IMO). Originally conceived as a tactical tool for collision avoidance, AIS has evolved into a strategic pillar for maritime Open Source Intelligence (OSINT). This evolution is driven by the integration of satellite-based receivers, allowing for the global monitoring of vessel movements even in remote oceanic regions where terrestrial signals cannot reach. However, the rising prevalence of dark vessel behaviors—where transponders are intentionally disabled or data is manipulated to evade detection—presents a significant challenge to maritime domain awareness, particularly in contested zones such as the Strait of Hormuz during periods of heightened geopolitical tension.

The Technical Architecture of the AIS Protocol and Data Link

The technical foundations of AIS are codified in Recommendation ITU-R M.1371, which delineates the physical, link, network, and transport layers of the system. Operating in the Very High Frequency (VHF) maritime mobile band, AIS utilizes two dedicated channels: AIS 1 (161.975 MHz) and AIS 2 (162.025 MHz). The system architecture relies on Time Division Multiple Access (TDMA) to manage the broadcast of information, dividing each one-minute frame into 2,250 time slots. Each slot lasts approximately 26.67 milliseconds, providing enough duration for a 256-bit packet to be transmitted at a rate of 9,600 bits per second. This synchronization ensures that multiple vessels within radio range can share the same frequency without significant signal overlap, a process managed through several distinct access schemes.

Self-Organized Time Division Multiple Access (SOTDMA) is the primary scheme for Class A shipborne stations. It allows the transponder to autonomously map the local radio environment, identifying free slots and pre-announcing its next transmission to reserve space in the frame. This decentralized coordination is what allows AIS to operate without a master controller, adapting dynamically to the density of local traffic. In contrast, Class B transceivers, typically found on smaller commercial and recreational vessels, use Carrier Sense Time Division Multiple Access (CSTDMA), which requires the device to listen for a clear slot before transmitting, effectively giving priority to Class A mandated vessels. Other schemes include Incremental TDMA (ITDMA) for pre-announcing transmissions during reporting interval changes and Fixed Access TDMA (FATDMA), which is primarily utilized by base stations to reserve slots for repetitive broadcasts like navigational corrections.

The physical layer is characterized by 9600-bit Gaussian Minimum Shift Keying (GMSK) frequency modulation. Data packets are encapsulated using High-Level Data Link Control (HDLC) protocols, ensuring data integrity through a 16-bit Frame Check Sequence (FCS). For OSINT analysts, the challenge lies in decoding the payload armoring, where the binary data is transformed into 6-bit ASCII characters for inclusion in NMEA-0183 sentences. This transformation is a critical step in the signal intelligence pipeline, as it converts raw radio waves into the structured data fields required for geospatial visualization and behavioral modeling.

Comprehensive Documentation of AIS Message Types 1 Through 27

The AIS protocol encompasses a full spectrum of 27 predefined message types, each serving a specific function within the maritime communication ecosystem. These messages provide a multi-layered view of a vessel's identity, physical characteristics, and dynamic movement.

Standard Position Reports and Dynamic Data

Messages 1, 2, and 3 constitute the standard position reports for Class A shipborne mobile equipment. These messages are broadcast frequently—ranging from every 2 seconds for high-speed vessels to every 3 minutes for those at anchor—to provide a real-time operational picture of the vessel’s course, speed, and heading.

Parameter	Bits	Description and Operational Significance
Message ID	6	Identifies the message as type 1, 2, or 3.
Repeat Indicator	2	Indicates how many times the message has been repeated (0-3).
User ID (MMSI)	30	Unique Maritime Mobile Service Identity for the station.
Navigational Status	4	Codes for status: 0=Under way using engine, 1=At anchor, 5=Moored, etc.
Rate of Turn (ROT)	8	Indicates turn rate from 0 to ±126 (708 degrees/min).
SOG	10	Speed over ground in 1/10 knot steps (0-102.2 knots).

Parameter	Bits	Description and Operational Significance
Position Accuracy	1	1=High (≤10m), 0=Low (>10m).
Longitude	28	Longitudinal position in 1/10,000 minute increments.
Latitude	27	Latitudinal position in 1/10,000 minute increments.
COG	12	Course over ground in 1/10 degree increments (0-359.9).
True Heading	9	Degrees (0-359). 511 indicates not available.
Time Stamp	6	UTC second when the report was generated (0-59).
Comm State	19	Diagnostic data for the TDMA slot allocation process.

The technical complexity of these messages is underscored by the Communication State field, which informs other stations of the vessel's future slot reservations. This field includes a synchronization state, a slot timeout indicating the number of frames until a new slot is selected, and a sub-message that provides additional diagnostic information.

Static, Voyage, and Management Messaging

Message 5 provides static and voyage-related data, which is typically broadcast every 6 minutes or upon request. Unlike position reports, Message 5 is a multi-slot message (occupying two slots) and contains information that remains relatively constant during a transit.

Parameter	Bits	Data Content and Analytical Value
IMO Number	30	Permanent identification number for the vessel.
Call Sign	42	Seven 6-bit ASCII characters for radio identification.

Parameter	Bits	Data Content and Analytical Value
Vessel Name	120	Maximum 20 characters in 6-bit ASCII.
Ship Type	8	Defines vessel category and hazardous cargo status.
Dimensions	30	Distances to bow, stern, port, and starboard from transponder.
ETA	20	Estimated Time of Arrival (MMDDHHMM UTC).
Draught	8	Current draught in 1/10 meter increments.
Destination	120	Intended port of arrival (20 characters).

Message 24 serves a similar purpose for Class B vessels, though it is split into two parts (A and B) to fit into single slots. Part A provides the vessel name, while Part B includes the ship type, call sign, dimensions, and a 42-bit Vendor ID which is critical for identifying the specific hardware manufacturer of the transponder.

Specialized and Binary Application Messages

Messages 6 and 8 are variable-length binary messages designed for addressed and broadcast communications, respectively. These "Application Specific Messages" (ASMs) allow authorities and commercial entities to transmit specialized data payloads, such as meteorological information, water levels, or port signals.

- Message 6 (Addressed Binary): Contains a source and destination MMSI, allowing for point-to-point data exchange up to 1,008 bits.
- Message 8 (Broadcast Binary): Allows a station to broadcast a binary payload to all receivers in range. These are often used for regional safety notices or pilotage instructions.
- Message 9 (SAR Aircraft Position): A specialized report for search and rescue aircraft, including a 12-bit altitude field capable of reporting heights up to 4,094 meters.
- Message 21 (AtoN Report): Transmitted by Aids to Navigation such as buoys and lighthouses. It indicates if the aid is "virtual" (transmitted from a shore station to mark a hazard without a physical structure) and if it has drifted "off-position".

Management of the AIS network itself is handled through messages like Message 15 (Interrogation), which allows a station to request specific message types from another vessel, and Message 16 (Assignment Mode), which enables a base station to force a mobile unit into a specific reporting interval. Message 22 (Channel Management) is used by competent authorities to assign alternate frequencies for a specific region, a tactic often used to manage bandwidth in congested ports.

Long-Range Satellite Tracking: Message 27

A critical development for global maritime surveillance is Message 27, the Long-Range AIS Broadcast Message. Traditional AIS messages are prone to collisions when received by satellites due to the large footprint of the satellite antenna and the propagation delay differences between vessels. Message 27 addresses this by compressing the position report into 96 bits and utilizing a specific reporting schedule.

Parameter	Bits	Description
Message ID	6	Constant 27.
User ID	30	Vessel MMSI.
Position	35	Longitude (18 bits) and Latitude (17 bits) in 1/10 min increments.
SOG	6	Speed in knots (0-62).
COG	9	Course in degrees (0-359).
Position Latency	1	Indicates if the reported position is older than 5 seconds.

Message 27 allows Class A and Class B+ vessels to be tracked globally without the high probability of signal blurring that affects standard position reports in high-traffic zones.

Identification and Mitigation of Dark Vessel Behaviors

In the context of maritime intelligence, a "dark vessel" refers to a ship that has intentionally disabled its AIS transponder or is manipulating its transmitted data to avoid monitoring. This behavior is frequently associated with illegal, unreported, and unregulated (IUU) fishing, sanctions evasion, smuggling, and unauthorized loitering near sensitive infrastructure such as subsea cables or wind farms.

Patterns of AIS Disabling and Spoofing

Intentional AIS disabling is often characterized by specific spatial and temporal markers. Analysts have identified that disabling events are significantly more frequent in proximity to Exclusive Economic Zone (EEZ) boundaries, particularly those that are contested or rich in resources but lack robust enforcement. For example, vessels may "go dark" just before entering a protected zone to hide illegal fishing activities, only to reactivate the transponder once they have exited the area.

AIS spoofing involves the transmission of false information, such as fake locations, identities, or nationalities. Advanced spoofing tactics include "track spoofing," where a vessel broadcasts a coherent but entirely fabricated trajectory, sometimes even showing the vessel loitering in a different ocean entirely while it is physically engaged in illicit activity. In conflict zones, spoofing is often a state-sponsored tactic used to degrade the utility of commercial tracking services for naval intelligence.

Non-Cooperative Visibility through Multi-Sensor Fusion

Since AIS is a cooperative system, its failure must be managed through maritime OSINT. This involves the fusion of AIS data with "non-cooperative" sensors that do not rely on the vessel's voluntary transmission.

Synthetic Aperture Radar (SAR) is the primary tool for detecting dark vessels. SAR sensors can penetrate cloud cover and operate during the night, detecting the metallic structures and wakes of ships. When a SAR detection is identified in an area where no AIS signals are present, the target is flagged as a dark vessel. This process of "AIS Correlation" searches for AIS messages within a certain radius (typically 1,500 meters) of the detection location.

Detection Phase	Technology Used	Analytical Objective
Target Acquisition	SAR / Optical Satellite	Locate physical presence of a vessel regardless of AIS status.
Identity Association	AIS Database Matching	Attempt to correlate the physical target with a known AIS broadcast.
Anomaly Detection	Machine Learning	Identify discrepancies in vessel dimensions or course vectors.

Detection Phase	Technology Used	Analytical Objective
Behavioral Analysis	Trajectory Modeling	Infer intent based on loitering or co-travel patterns with known vessels.

Deep learning models, such as YOLOv11-OBB (Oriented Bounding Box), are increasingly used to refine this process. By estimating a vessel's course and dimensions directly from satellite imagery, analysts can verify if the physical characteristics of the target match the static data reported in AIS Message 5. Deviations in length or beam of more than 10-15% are strong indicators of identity deception or cloning.

Case Study: Maritime Intelligence in the Strait of Hormuz Conflict (2026)

The Strait of Hormuz represents a critical failure point for transparent maritime tracking during military escalations. Following the initiation of "Operation Epic Fury" on February 28, 2026, the maritime intelligence landscape shifted from routine commerce monitoring to active signal warfare analysis.

Operational Collapse and Traffic Redistribution

The conflict resulted in an immediate 80-94% reduction in traceable transits through the Strait. While the waterway was not physically blocked, a "financial blockade" emerged as war-risk insurance premiums surged and major global operators suspended transits due to the physical danger of vessel strikes.

Incident Tracking	Vessel Name	IMO Number	Signal Status and Outcome
Engine Explosion	*MKD Vyom*	9284386	Signal lost after strike; one fatality reported.
Fire at Sea	*Skylight*	9330020	Intermittent AIS before strike; sanctioned vessel.
GPS Spoofing	*Sea La Donna*	9380532	Reported location inconsistent with imagery.
Evacuation / Turn	*KHK Empress*	9534963	Westbound transit reversed abruptly at 16 knots.

The withdrawal of reputable owners left the corridor occupied primarily by high-risk, sanctioned, or "shadow fleet" operators who frequently disabled their AIS transponders to navigate "dark" and reduce exposure to targeting.

Electronic Warfare and Navigation Interference

A hallmark of the 2026 crisis was the mass degradation of GPS reliability across the Middle East Gulf. More than 1,100 vessels were tracked experiencing significant navigation interference in a 24-hour window. This interference manifest as "AIS diversion," where vessel positions were spoofed to inland locations such as airports or nuclear facilities, creating false positive compliance breaches for marine service providers.

Vessels in the region adapted by using AIS as an active signaling mechanism rather than just a navigational tool. Commercial operators broadcasted neutral messages in their destination or name fields, such as "NO IL LINK" or "ALL MUSLIMS ON BOARD," in an attempt to signal their lack of affiliation with combatant states and mitigate the risk of seizure or attack. This use of the AIS static data fields for defensive psychological signaling is an emerging trend in gray-zone maritime conflict.

API Schemas for Global Vessel and Cargo Tracking

For the signal intelligence professional, accessing raw AIS data at scale requires integration with commercial API services. MarineTraffic and AISHub are the two primary providers, each with distinct schemas and access models.

MarineTraffic AIS API (Version 9)

The MarineTraffic Vessel Positions API provides the most granular data for global tracking, supporting both terrestrial and satellite-received signals. The response is structured around a DATA array of objects and a METADATA object for pagination.

JSON

```
{
  "DATA":,
  "METADATA": {
    "CURSOR": "abcdef123",
    "DATE_FROM": "2023-11-20 15:57:00"
  }
}
```

The DSRC parameter is essential for conflict zone analysis, as it indicates the data source: TER for terrestrial antennas (near-shore), SAT for satellite (open-ocean), or ROAM for

roaming receivers. The timespan parameter allows the API to return the latest known position within a window of up to 24 hours, which is vital for monitoring vessels that have recently gone dark.

AISHub API and Data Sharing Schema

AISHub provides a web service for raw and human-readable AIS data. The API is credit-based but requires users to share their own AIS feed to maintain access to the aggregated global stream.

Parameter	Function	Operational Usage
username	Auth	Assigned after joining the sharing network.
format	Encoding	0 for raw AIS encoding; 1 for Human-readable JSON/XML.
latmin/max	Spatial	Defines the bounding box for the query.
interval	Temporal	Maximum age of positions in minutes.

AISHub's schema includes unique fields such as SID (Station ID), which identifies which specific terrestrial station received the message, allowing analysts to map the coverage reliability of the terrestrial network in regions like the Strait of Hormuz.

Container and Logistics Intelligence

Advanced tracking APIs, such as the MarineTraffic Container Intelligence V2, integrate AIS positions with cargo-level data. This allows for the tracking of specific shipments on vessels even if the vessel's AIS is intermittent. The schema provides "milestones" for the supply chain, including terminal congestion indices, average waiting times, and carrier performance metrics, which are essential for predicting the economic impact of maritime disruptions.

Architectural Orchestration of a Geospatial Command Center

The integration of these signal streams into an operational picture requires a high-performance rendering environment capable of fusing multi-modal data onto a geodetic ellipsoid. The standard for this in 2026 is the spatial orchestration of a "God's Eye" view using CesiumJS and WebGPU integration.

Visual Foundation and Sensor Layering

A global command center must stream petabytes of geographical data with sub-meter precision. The use of WebGPU allows for the efficient processing of massive point clouds and real-time Gaussian splats, which characterize modern digital twins of physical reality. The system overlays live "leaky" data streams—aviation (ADS-B), maritime (AIS), and orbital (TLE)—onto a textured 3D mesh such as Google Photorealistic 3D Tiles.

For maritime tracking, the use of "4D paths" in the CZML (Cesium Language) standard is essential. These paths allow an operator to "scrub" through a vessel's trajectory over time, identifying route deviations or loitering patterns that precede a dark activity event. orientation is automatically computed using velocity orientation properties, ensuring that 3D vessel models are oriented correctly according to their heading and course over ground.

Agentic Reasoning and Signal Analysis

The complexity of conflict zone monitoring necessitates an agentic layer to function as the command center's brain. This layer consists of scraper agents and reasoning agents built on large language models (LLMs) such as Gemini Pro.

- **Scraper Agents: These monitor unstructured sources, including Telegram channels and live news feeds, transforming raw text into structured GeoJSON pings.**
- **Reasoning Agents: These perform data fusion to identify complex patterns. For instance, an automated alert for a potential naval incident in the Strait of Hormuz might be triggered by the confluence of GPS jamming spikes, commercial flight diversions, and a dark vessel detection from a SAR satellite pass.**

Advanced localization algorithms, such as the "Radio Horizon" method, can be implemented within this layer to pinpoint ground-based jammers. By identifying the moment an aircraft's Navigation Integrity Category (NIC) declines and calculating the intersection density of radio horizons from multiple aircraft, the system can triangulate the most likely coordinates of the electronic warfare source.

Summary of Strategic Maritime Domain Awareness

The Automatic Identification System has transitioned from a local safety aid to a global strategic intelligence resource. The protocol's 27 message types provide a robust but spoofable framework for maritime transparency. Effective domain awareness in the modern era requires a transition from reactive monitoring to proactive, multi-sensor surveillance. This involves the systematic use of SAR/Optical fusion to unmask dark vessels, the integration of real-time API feeds for supply chain visibility, and the orchestration of agentic geospatial environments for tactical reasoning. In regions like the Strait of Hormuz, where the signal environment is actively contested, these advanced OSINT and SIGINT methodologies are the only means of maintaining an accurate operational picture amidst the chaos of electronic warfare and strategic deception.